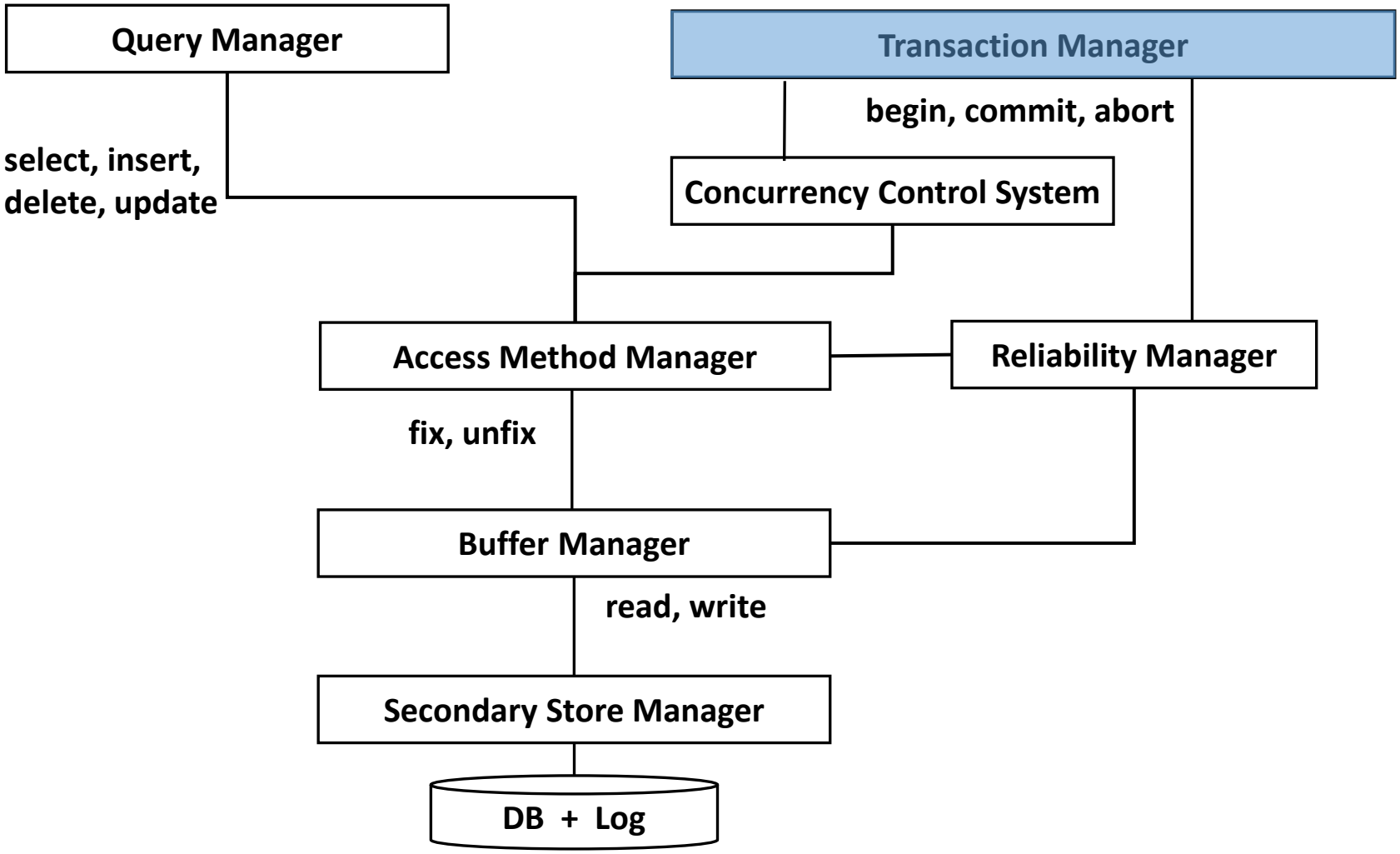
Data bases 2

Transactional Systems

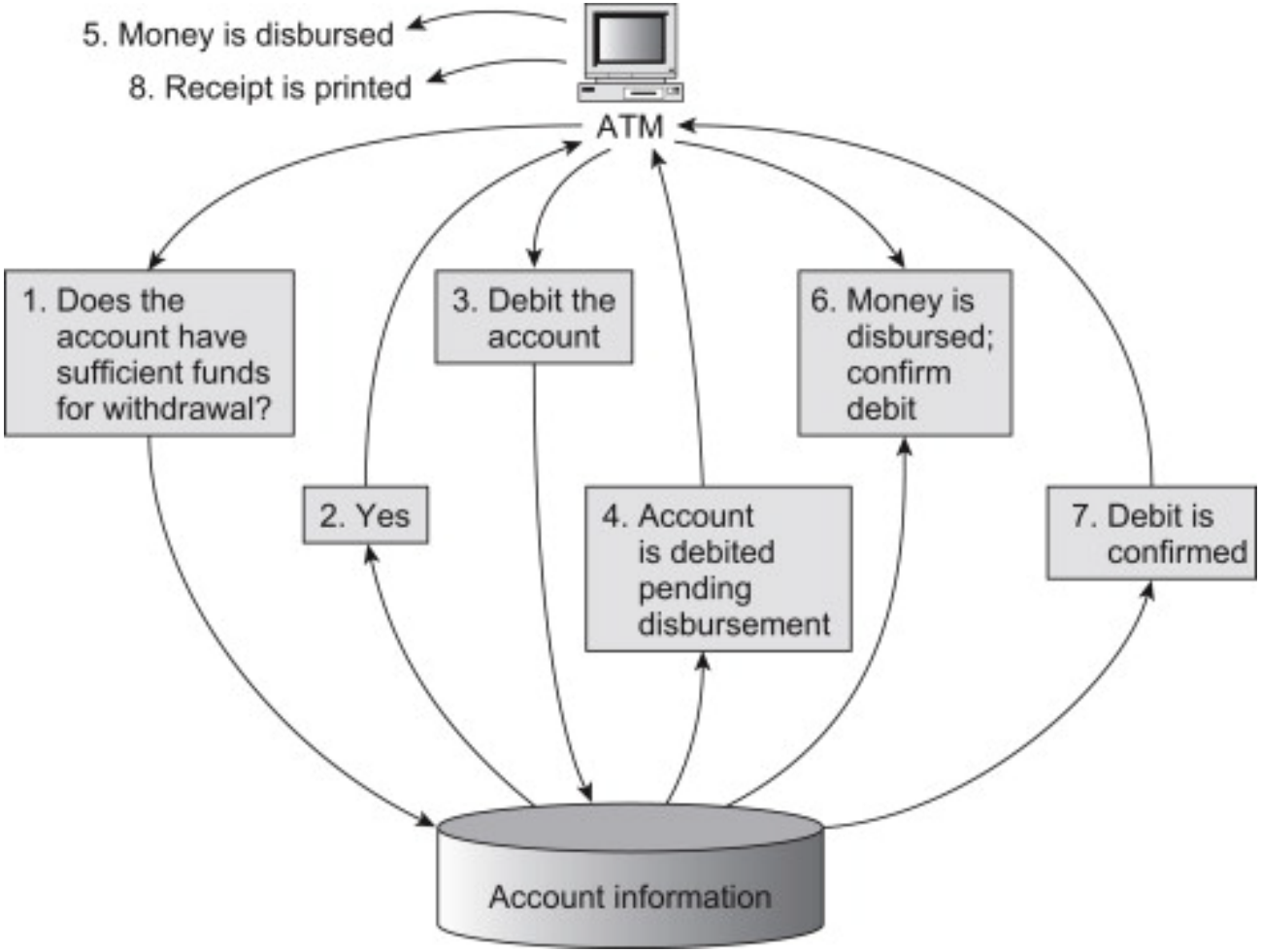
In this lecture

- The architecture of the DDBMS
- Introduction to the concept of transaction
- Properties of a transaction
- Modules of the DBMS responsible of guaranteeing such properties

The architecture of the DBMS



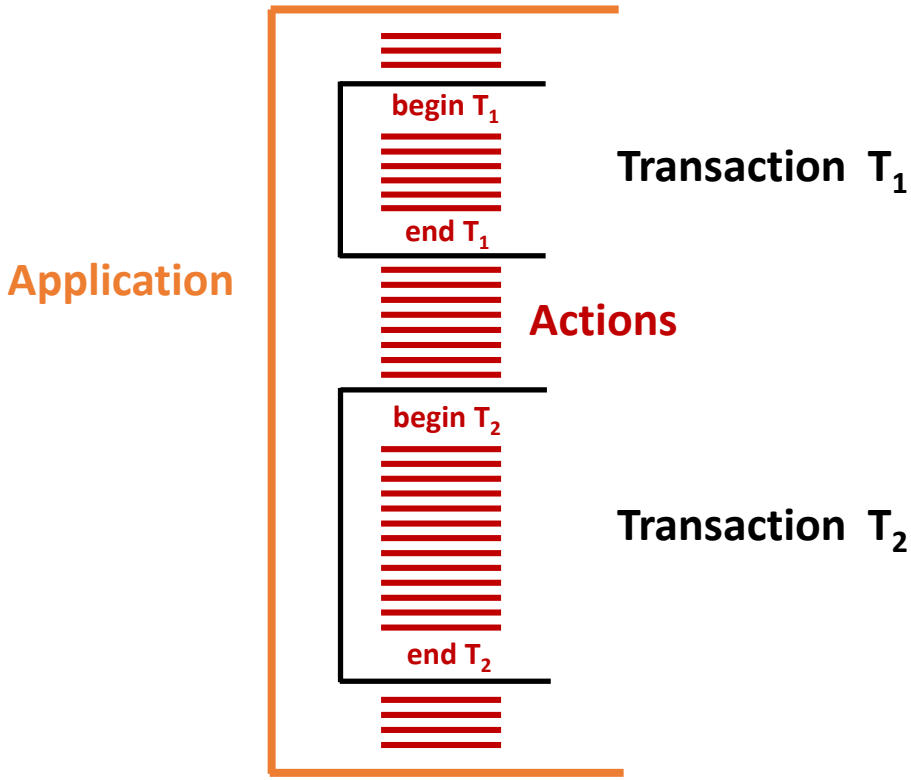
The need of transactions: ATM example



Definition of Transaction

- An elementary, atomic unit of work performed by an application
- Each transaction is conceptually encapsulated within two commands:
  - begin transaction
  - end transaction
- Within a transaction, one of the commands below is executed (exactly once) to signal the end of the transaction:
  - commit-work (commit)
  - rollback-work (abort)
- Transactional System (OLTP): a system that supports the execution of transactions on behalf of concurrent applications

Difference between Application and Transaction



Transactions: bank transfer

```
begin transaction;

update Account
set Balance = Balance + 10
where AccNum = 12202;

update Account
set Balance = Balance - 10
where AccNum = 42177;

commit-work; // end transaction
```

Transactions: bank transfer upon condition

```
begin transaction;
update Account
set Balance = Balance + 10 where AccNum = 12202;

update Account
set Balance = Balance - 10 where AccNum = 42177;

select Balance into A from Account
where AccNum = 42177;

if ( A >= 0 ) then
commit-work; // end transaction with success
else
rollback-work; // end transaction with failure
```

ACID Properties of Transactions

- A transaction is a unit of work characterized by the following properties:
  - **A**tomicity
  - **C**onsistency
  - **I**solation
  - **D**urability

Atomicity

- A transaction is an indivisible unit of execution
  - Either all the operations in the transaction are executed or none is executed
- In the case of the bank transfer, the execution of a single update statement would be disastrous
- The time in which COMMIT is executed marks the instant in which the transaction ends successfully:
  - An error before should cause the rollback of the work
  - An error afterwards should not alter the effect of the transaction
- The ROLLBACK of the work performed can be caused
  - By the application with a ROLLBACK statement
  - By the DBMS, for example for the violation of integrity constraints or for concurrency management
- In case of a rollback, the work performed must be undone, bringing the database to the state it had before the start of the transaction
- It is the application's responsibility to decide whether an aborted transaction must be redone or not

Consistency

- A transaction must satisfy the DB integrity constraints
  - if the initial state S_0 is consistent
  - then the final state S_f is also consistent
  - This is not necessarily true for the intermediate states S_i
- Example
  - The sum of the worked hours per task should equal the planned work hours of the project
    - If the constraint holds before the transaction it must hold also after its execution
    - The constraint can be temporarily violated during the execution of the transaction, e.g., when shifting work from a task to another one, but must be satisfied at the end

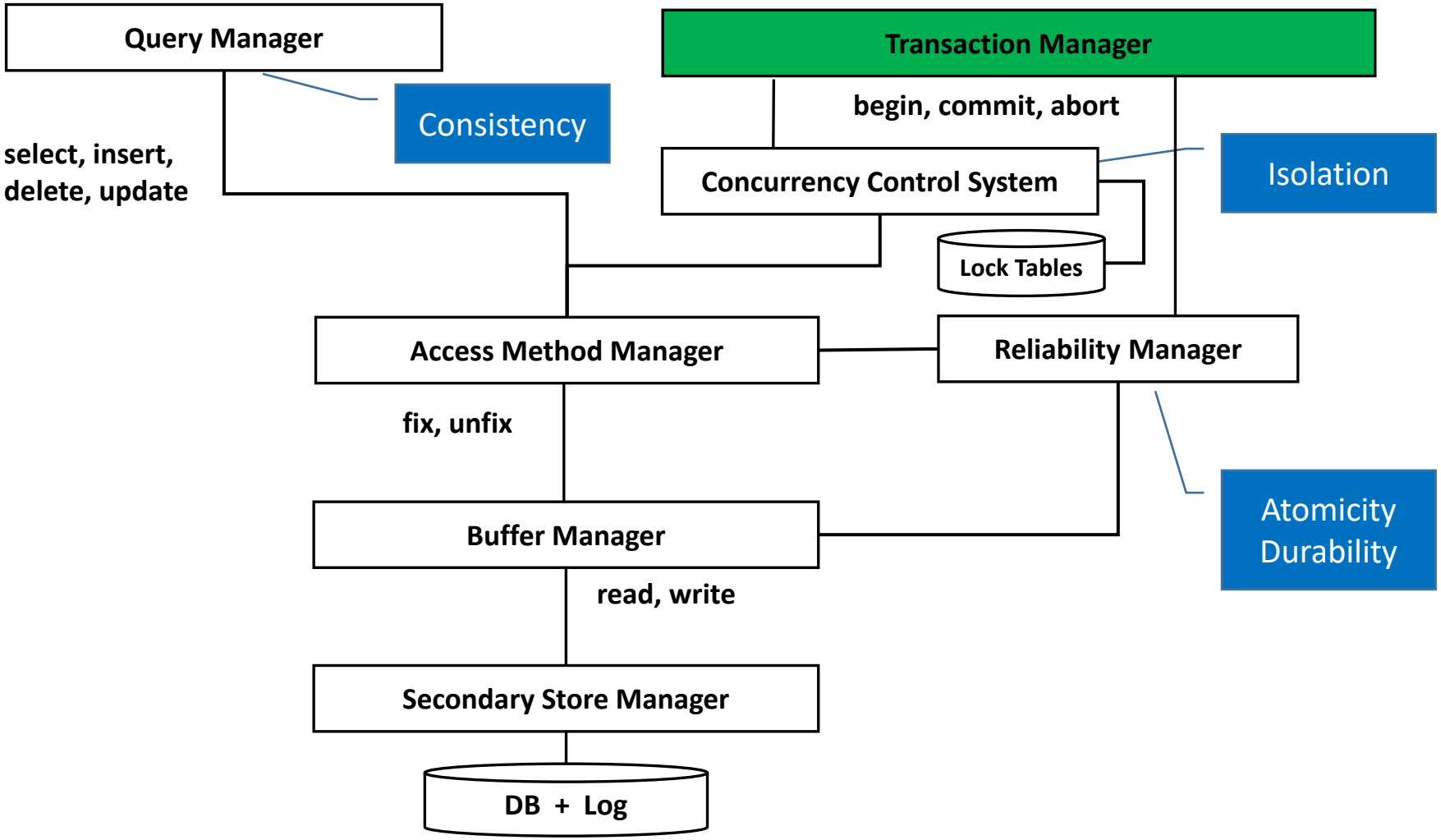
Isolation

- The execution of a transaction must be independent of the concurrent execution of other transactions
- In particular, the concurrent execution of a number of transactions **must produce the same result as the execution of the same transactions in a sequence**
- E.g., the concurrent execution of transactions T1 and T2 must produce the same results that can be obtained by executing one of these sequences
  - T1,T2
  - T2,T1
- Isolation impacts performance and trade-offs can be defined between isolation and performance

Durability

- The effect of a transaction that has successfully committed will last "forever"
  - Independently of any system fault
  - Sounds obvious… but every DBMS manipulates data in main memory

ACID properties & DBMS mechanisms

- Atomicity
  - Abort-rollback-restart, commit protocols
  - Reliability Manager
- Consistency
  - Integrity checking of the DBMS
  - Integrity Control System at query execution time (with the support of the DDL compiler)
- Isolation
  - Concurrency control
  - Concurrency Control System
- Durability
  - Recovery management
  - Reliability Manager

Transactions & DBMS architecture